Kassapa Gamagedara

Professional Summery

- PhD candidate in Physics with deep expertise in tribo-electrochemistry, CMP slurry optimization, and surface/interface science.
- Proven track record in research, process optimization, and lab management.
- Recognized with awards and publications in peer-reviewed journals.
- Seeking to apply scientific problem-solving and technical leadership to drive innovation in semiconductor industry.

Key Skills

- CMP process development, slurry formulation, corrosion inhibition
- Electroanalytical methods: OCP, PDP, EIS, impedance modeling
- Experimental tools: CMP polishers, Potentiostat + FRA, AFM, SEM, glove box systems
- Data analysis: OriginLab, ZPlot/ZView, ZSimpWin, VersaStudio, Excel
- Lab management, safety compliance, training & mentoring

Research and Technical Experience

Graduate Researcher – CMP & Surface Science, Clarkson University | 2019–Present

- Designed and optimized CMP slurries for Cu, Co, Mo, and stainless steel.
- Investigated galvanic corrosion suppression strategies in Co/Cu systems.
- Developed post-CMP cleaning achieving 97% oxide residue removal using tartrate brush scrubbing.
- Applied electroanalytical diagnostics (OCP, PDP, EIS) under polishing.
- Managed lab operations: safety compliance, equipment upkeep, budgeting, training of junior researchers.

Graduate Teaching Assistant – Clarkson University | 2019-present

- Served as a Teaching Assistant for PH131/PH132 labs (mechanics, electromagnetism), developing skills in instruction, mentoring, and technical communication

Selected Publications

Gamagedara, K.U.; Roy, D. Tribo-Electrochemical Considerations for Assessing Galvanic Corrosion in CMP, Electrochem, 2025.

Santefort, D.R.; Gamagedara, K.U.; Roy, D. Stainless Steel CMP Using Citrate Buffer, Materials, 2025.

Gamagedara, K.U.; Roy, D.; Santefort, D.R. Experimental Strategies for Tribo-Electrochemical CMP Studies, Lubricants, 2024.

Santefort, D.; Gamagedara, K.U.; Roy, D. Tribo-Electroanalytical Evaluation of CMP Slurries and Post-CMP Cleaning Solutions, ICPT Proceedings, 2022.

Highlights

- Ph.D. Physics
- 6 publications
- CMP slurries & post CMP cleaning
- Electrochemical deposition
- Teamwork and collaboration
- Awards: Outstanding Grad Student (2025), Best poster (2022)

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Awards and recognition

Outstanding Graduate Student Award Coulter School of Engineering, Clarkson University (2025)

1st Place, Best Graduate Poster in Electrochemistry, Clarkson Research & Project Showcase (2022)

Education

Expected 12/2025
Ph.D. in Physics
Clarkson University, Potsdam, NY
2021–Present

M.Sc. in Physics
Clarkson University, Potsdam, NY
2019–2021

M.Sc. in Physics of Materials
University of Peradeniya, Sri Lanka
2016–2018

B.Sc. in Physics, Mathematics & Statistics
University of Peradeniya, Sri Lanka
2013–2015

Research Projects & Applications

My doctoral research integrates tribo-electrochemistry and process optimization in Chemical Mechanical Planarization (CMP), focusing on electrochemical diagnostics, surface metrology, and slurry development for advanced metals and alloys relevant to semiconductor manufacturing.

Project 1: Tribo-Electroanalytical Evaluation of CMP Slurries

Publication: Tribo-Electroanalytical Evaluation of CMP Slurries and Post-CMP Cleaning Solutions (ICPT Proceedings, 2022)

Pioneered combined tribological-electrochemical process diagnostics for Mo/Cu systems. Used polish-hold techniques and in-situ potential tracking to link frictional and electrochemical responses to removal behavior.

Key skills: Demonstrates real-time process monitoring, data acquisition and analysis, and cross-disciplinary process understanding, foundational for smart manufacturing and CMP tool control.

Project 2: Tribo-Electrochemical Mechanisms in CMP

Publication: Experimental Strategies for Studying Tribo-Electrochemical Aspects of CMP (Lubricants, 2024)

Developed a tribo-electrochemical framework to evaluate the coupling between mechanical abrasion and electrochemical reactions during CMP. Built integrated test cells and optimized experimental repeatability through signal-noise minimization and process control methods. Demonstrated strong control over OCP and PDP responses under load.

Key skills: Demonstrates process design, experimental optimization (DOE), and data-driven validation and mirroring fab-level process-development.

Project 3: Chemical Selectivity in Mo and Cu CMP

Publication: Mechanisms of Chemically Promoted Material Removal for Mo and Cu CMP (Materials, 2024)

Conducted comparative CMP studies for Mo and Cu under identical citrate chemistries to understand chemical–mechanical selectivity. Used electrochemical diagnostics (OCP, PDP, EIS) to correlate dissolution behavior with removal rates and pH-dependent film formation.

Key skills: Reflects process-parameter optimization, selectivity control, and chemical compatibility analysis vital for BEOL interconnect development and multi-material integration.

Project 4: Stainless-Steel CMP for Flexible Electronics

Publication: Tribo-Electrochemical Mechanism of Material Removal in Stainless-Steel CMP (Materials, 2025)

Investigated stainless-steel 316/316L CMP to enable low-defect, flexible-substrate applications. Conducted systematic pH and pressure studies to map the passive–active transition and correlate electrochemical kinetics with MRR and roughness.

Key skills: Integrates process scalability, material characterization, and data correlation, translating to skills in substrate engineering and advanced-packaging process development.

Project 5: Galvanic Corrosion in Multi-Metal CMP

Publication: Tribo-Electrochemical Considerations for Assessing Galvanic Corrosion in CMP (Electrochem, 2025)

Explored galvanic coupling between Cu and Co during simultaneous CMP to identify potential-driven defect formation. Modeled ion and electron transfer under slurry exposure and optimized pH/inhibitor concentration to minimize galvanic currents.

Key skills: Applies root-cause analysis, chemical-process control, and failure-mitigation strategies; key competencies for barrier-liner process reliability.

Project 6: Post-CMP Cleaning Optimization

Publication: Tribo-Electrochemical Characterization of Brush-Scrubbed Post-CMP Cleaning (Lubricants, 2025)

Designed a tartrate-based brush-scrubbing process for Cu post-CMP cleaning, integrating electrochemical feedback for real-time oxide-removal tracking. Achieved 97% oxide clearance with improved passivation and surface stability confirmed by impedance and contact-angle analysis.

Key skills: Demonstrates defectivity reduction, surface-integrity control, and wet-clean optimization directly aligned with yield-enhancement and contamination-control engineering.

My research experiences combine scientific precision with practical process insight, bridging the gap between tribo-electrochemical research and semiconductor process development. I bring strengths in process optimization, data analysis, and cross-functional collaboration, ready to contribute to materials R&D and advanced manufacturing innovation.